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The Erdős $n^2/25$ max-cut conjecture for small multiples of five, via a per-root-MaxCut envelope and blow-up integrality

Alper Ferudun*

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Abstract

Erdős conjectured that every triangle-free graph on N vertices can be made bipartite by deleting at most $N^2/25$ edges; the bound would be sharp, attained by the balanced blow-up $C_5[N/5]$. Writing $\beta(G)$ for the minimum number of edges whose deletion makes G bipartite and $a(N) = \max\{\beta(G) : G \text{ triangle-free on } N \text{ vertices}\}$, the conjecture is $a(N) \leq N^2/25$, and for $N = 5n$ it reads $a(5n) \leq n^2$. Balogh, Clemen and Lidícký proved it for large N in the two density tails (edge density at most 0.2486 or at least 0.3197) and proved the global bound $a(N) \leq N^2/23.5$; the medium-density band remains open. We prove

$$a(5n) = n^2 \qquad \text{for every } 1 \leq n \leq 40, \quad \text{i.e. } N \in \{5, 10, \ldots, 200\}.$$

The proof is computer-assisted and combines three ingredients. (i) A *per-root-MaxCut envelope*: for the 107 triangle-free 7-root types, the mean over types of the best per-type cut is an upper bound $d_{\mathrm{mono}}(W) \leq U_7(W)$ that is *tight* at the C_5-blow-up. (ii) An order-10 flag-algebra certificate—the per-root-MaxCut rows at 7 and 8 roots together with rooted-Horn cuts and a manifestly-PSD moment block—bounds the envelope on the medium band, $U_7(W) \leq \frac{2}{25} + \delta$ with an explicit rational $\delta \approx 4.8558 \times 10^{-5}$, for every triangle-free graphon W of edge density in $[0.2486, 0.3197]$. (iii) The blow-up identity $\beta(G[t]) = t^2\beta(G)$ plus integrality of β turns this into $\beta(G) \leq n^2 + \frac{25}{2}n^2\delta$ for any $5n$-vertex band-density G, and $\frac{25}{2}n^2\delta < 1$ for $n \leq 40$; the two density tails are handled by the Balogh–Clemen–Lidícký bounds, transferred to finite N by the same blow-up. The envelope bound $d_{\mathrm{mono}} \leq U_7$ is a genuine graphon upper bound (each per-root rule is one global 2-colouring), the certificate is verified in exact rational arithmetic, the moment positivity is Razborov's flag-algebra theorem exhibited as an exact Gram factorization, and the bound is cross-checked against brute-force max-cut on all triangle-free graphs of order at most 12. The same envelope at orders 9 and 10 provably does not reach the constant needed for larger n; we explain why, and locate the all-n conjecture at a single self-tight obstruction.

1 Introduction

For a graph G let $\beta(G)$ denote the *bipartization number*: the minimum number of edges whose removal leaves a bipartite graph, equivalently $e(G) - \mathrm{mc}(G)$ where $\mathrm{mc}(G)$ is the size of a maximum cut. Erdős asked how large $\beta(G)$ can be for a triangle-free graph; he conjectured (see [3]) that

$$\beta(G) \leq \frac{N^2}{25} \qquad \text{for every triangle-free } G \text{ on } N \text{ vertices,} \tag{1}$$

*`alper@mercurycodelab.com`. The envelope certificate, its exact-arithmetic verifier, and the brute-force ground-truth checker accompany this note as ancillary files; development history at `github.com/AlperTheKing`.

and that this is sharp. Sharpness is witnessed by the balanced blow-up of the 5-cycle: $C_5[m]$ (five independent sets of size m, complete bipartite between cyclically consecutive sets) is triangle-free on $N = 5m$ vertices with $\beta(C_5[m]) = m^2 = N^2/25$. Setting $a(N) = \max\{\beta(G) : G$ triangle-free on N vertices$\}$, (1) is the statement $a(N) \le N^2/25$; on multiples of five it reads $a(5n) \le n^2$, and the blow-up lower bound gives $a(5n) \ge n^2$, so the conjecture on multiples of five is exactly

$$a(5n) = n^2.$$

This is problem #23 in Bloom's database of Erdős problems [3].

The strongest progress to date is due to Balogh, Clemen and Lidícký [2], using the flag-algebra method of Razborov [5]. They proved the global bound $a(N) \le N^2/23.5$ and established the sharp bound (1) in the two density tails: for N large, $\beta(G) \le N^2/25$ whenever the edge density of G is at most 0.2486 or at least 0.3197 (their Theorem 1.3). The *medium-density band* $(0.2486, 0.3197)$ remains open, and is where the difficulty of (1) now resides; the extremal density $2/5$ of $C_5[m]$ lies in the high tail.

Result. We resolve the conjecture on the first eleven multiples of five.

Theorem 1.1. *$a(5n) = n^2$ for every integer $1 \le n \le 40$; equivalently, every triangle-free graph on N vertices with $N \in \{5, 10, 15, \ldots, 200\}$ can be made bipartite by deleting at most $N^2/25$ edges, and this is sharp.*

In particular $a(25) = 25$, $a(30) = 36$, …, $a(200) = 1600$. The sequence $a(N)$ is catalogued as OEIS A389646 [1] and has been determined exactly only up to $N = 23$ by direct enumeration. Every multiple of five in that known range agrees with Theorem 1.1: the catalogued values $a(5) = 1$, $a(10) = 4$, $a(15) = 9$, $a(20) = 16$ are exactly $1^2, 2^2, 3^2, 4^2$, an independent brute-force confirmation on each case where the truth is already known. Theorem 1.1 adds the multiples of five from 25 to 200 ($a(25) = 25$ through $a(200) = 1600$), all beyond the exhaustively determined range $N \le 23$ and hence new. The closely related extremal question — how many edges *must* be deleted — was studied by Erdős, Győri and Simonovits [4].

Method and honesty. The proof is *computer-assisted*. Its mechanism is to upper-bound the bipartization density of a triangle-free graphon by a *per-root-MaxCut envelope* (Section 3) that is tight at the extremal C_5-blow-up, certify the envelope on the medium band with an order-10 flag-algebra linear program (Theorem 3.2, certified value $\delta \approx 4.8558 \times 10^{-5}$), and convert this approximate graphon bound into an exact finite statement by integrality of β under blow-up (Section 2); since $\frac{25}{2}n^2\delta < 1$ only for $n \le 40$, the closure stops at $n = 40$. The density tails are cited from [2] (Section 4); the assembly is in Section 5; the verification is in Section 6.

We separate known from new. The conjecture and its sharp example are Erdős's; the tail bounds, the global $N^2/23.5$ bound, and the flag-algebra framework for this problem are due to [2, 5]. New here are: the per-root-MaxCut envelope and its order-10 certificate, the blow-up integrality reduction, and the resulting finite-range Theorem 1.1. Theorem 1.1 does *not* resolve (1): for $n \ge 12$ the certificate margin exceeds the integrality gap, and (Section 7) the envelope provably cannot be sharpened enough at orders 9 or 10; the all-n conjecture sits at a single self-tight obstruction.

2 The blow-up integrality reduction

For a graph G and integer $t \geq 1$, the *t-blow-up* $G[t]$ replaces each vertex by an independent set of size t and each edge by a complete bipartite graph. Two facts about bipartization under blow-up drive everything.

Lemma 2.1. *For every graph G and every $t \geq 1$, $\beta(G[t]) = t^2\,\beta(G)$.*

Proof. We show $\mathrm{mc}(G[t]) = t^2\,\mathrm{mc}(G)$; since $e(G[t]) = t^2 e(G)$ and $\beta = e - \mathrm{mc}$, the claim follows. The t copies of a vertex are pairwise non-adjacent twins with identical neighbourhoods, so any cut value depends only on *how many* copies of each class lie on each side, not on which. Hence the cut value of $G[t]$ is a multilinear function of the fractional assignment $x_v \in [0,1]$ giving the proportion of the blown-up class of v placed on one side: an edge uv contributes $t^2\big(x_u(1-x_v) + x_v(1-x_u)\big)$. A multilinear function on the cube $[0,1]^{V(G)}$ attains its maximum at a vertex of the cube, i.e. at an integral $x \in \{0,1\}^{V(G)}$, which is an ordinary cut of G scaled by t^2. Hence the maximum fractional cut equals $t^2\,\mathrm{mc}(G)$ and there is no fractional–integral gap, so $\mathrm{mc}(G[t]) = t^2\,\mathrm{mc}(G)$. □

Graphon density. Normalising by $\binom{N}{2} \to N^2/2$, define the *monochromatic-pair density* of G on N vertices as $d_{\mathrm{mono}}(G) = \beta(G)/(N^2/2) = 2\beta(G)/N^2$. By Lemma 2.1 this is blow-up invariant: $d_{\mathrm{mono}}(G[t]) = 2t^2\beta(G)/(Nt)^2 = 2\beta(G)/N^2 = d_{\mathrm{mono}}(G)$ for every t. Let W_G be the step-graphon of G (equivalently, of every $G[t]$). Since the sequence $d_{\mathrm{mono}}(G[t])$ is constant in t, its limit — the graphon value $d_{\mathrm{mono}}(W_G)$ — equals $2\beta(G)/N^2$ *exactly*, with no $O(1/t)$ residual; this is the only point at which the finite G enters the graphon bound. Thus any valid upper bound $d_{\mathrm{mono}}(W) \leq \frac{2}{25} + \delta$ over triangle-free graphons of a given edge density applies verbatim to every finite G of that density:

$$\beta(G) = \tfrac{N^2}{2}\, d_{\mathrm{mono}}(G) \leq \tfrac{N^2}{2}\Big(\tfrac{2}{25} + \delta\Big) = \frac{N^2}{25} + \frac{N^2\delta}{2}. \tag{2}$$

Proposition 2.2. *Suppose $d_{\mathrm{mono}}(W) \leq \frac{2}{25} + \delta$ for every triangle-free graphon W whose edge density lies in a band B. Then for every triangle-free graph G on $N = 5n$ vertices whose edge density lies in B,*

$$\beta(G) \leq n^2 + \frac{25}{2}\, n^2\, \delta.$$

In particular, if $\frac{25}{2} n^2 \delta < 1$ then $\beta(G) \leq n^2$, because $\beta(G)$ is a nonnegative integer.

Proof. Apply (2) with $N = 5n$: $N^2/25 = 25n^2/25 = n^2$ and $N^2\delta/2 = 25n^2\delta/2$. The integrality rounding is immediate. □

With the certified value $\delta \approx 4.8558 \times 10^{-5}$ (Theorem 3.2), the threshold $\frac{25}{2} n^2 \delta < 1$ holds for $n \leq 40$: $n = 40$ requires $\delta < 2/(25 \cdot 1600) = 5.0000 \times 10^{-5}$, which holds with a 2.88% margin, while $n = 41$ requires $\delta < 4.7591 \times 10^{-5}$, which fails. This is the source of the bound $n \leq 40$ in Theorem 1.1.

3 The per-root-MaxCut envelope

The certificate bounds d_{mono} not by a fixed cut but by an *envelope* of per-root maximum cuts, which is what lets it be tight at the extremal C_5-blow-up. Let $\mathcal{T}_7$ be the set of the 107 isomorphism classes of triangle-free graphs on 7 vertices (the *root types*). Sampling an ordered 7-tuple of vertices, we map its induced graph to its canonical representative $\sigma \in \mathcal{T}_7$ by a fixed canonical isomorphism (the lexicographically least), read off each remaining vertex's profile in the resulting canonical root coordinates, and write $p_\sigma(W)$ for the density of ordered 7-tuples whose induced graph is isomorphic to σ (automorphisms and root embeddings accounted for in the density), so $\sum_\sigma p_\sigma(W) = 1$. Fix σ and a set of 7 roots of type σ. Each other vertex v has a *profile* $\pi_\sigma(v) \in \{0,1\}^7$ (its adjacencies to the roots); a *rule* $c\colon \{0,1\}^7 \to \{0,1\}$ assigns a side to each profile, hence a 2-colouring of the non-root vertices. Write $g_{\sigma,c}(W) = p_\sigma(W)\big(\mathbb{E}[C \mid \sigma, c] - \frac{2}{25}\big)$ for the (order-≤ 9) flag density of the resulting monochromatic-pair deficit, where $\mathbb{E}[C \mid \sigma, c]$ is the conditional monochromatic-pair density of the colouring. Define the *per-root-MaxCut envelope*

$$U_7(W) \;:=\; \sum_{\sigma \in \mathcal{T}_7} p_\sigma(W) \, \min_c \, \mathbb{E}[C \mid \sigma, c] \;=\; \frac{2}{25} + \sum_{\sigma \in \mathcal{T}_7} \min_c \, g_{\sigma,c}(W),$$

the last equality using $\sum_\sigma p_\sigma = 1$ and $p_\sigma \geq 0$. For each σ, $\min_c \mathbb{E}[C \mid \sigma, c]$ is the minimum monochromatic density over rules, i.e. a *maximum cut of the profile graph* of σ.

Lemma 3.1 (envelope soundness)**.** *For every triangle-free $\{0,1\}$-valued step graphon W — in particular the blow-up graphon W_G of any finite triangle-free graph G — one has $d_{\mathrm{mono}}(W) \leq U_7(W)$, with equality at the C_5-blow-up.*

Proof. On a $\{0,1\}$ step graphon every vertex v has a genuine adjacency bit to each root, so the profile $\pi_\sigma(v) \in \{0,1\}^7$ is well defined and the rule below yields an honest 2-colouring. Sample 7 roots R as i.i.d. points of W; condition on R inducing $\sigma \in \mathcal{T}_7$ and let $c^*(\sigma) = \arg\min_c \mathbb{E}[C \mid \sigma, c]$. Colour every other vertex v by $c^*(\sigma)\big(\pi_\sigma(v)\big)$. Since the roots are a null set, this is a genuine measurable 2-colouring φ_R of W, so its monochromatic-pair density is at least the minimum over all colourings, $\mathrm{mono}(\varphi_R) \geq d_{\mathrm{mono}}(W)$. Taking the expectation over R,

$$d_{\mathrm{mono}}(W) \;\leq\; \mathbb{E}_R\big[\mathrm{mono}(\varphi_R)\big] \;=\; \sum_\sigma p_\sigma(W) \, \mathbb{E}[C \mid \sigma, c^*(\sigma)] \;=\; U_7(W).$$

At $W = C_5[N/5]$ the optimal 5-class colouring is realised by $c^*(\sigma)$ for the relevant root types, giving $U_7 = d_{\mathrm{mono}} = 2/25$ (verified in Section 6). □

The $\{0,1\}$ hypothesis is all the finite theorem needs: every finite G enters only through its blow-up graphon W_G (Section 2), a $\{0,1\}$ step graphon. (For a general graphon the same bound holds by independently rounding the fractional profile, but we do not use it.) This is the decisive structural gain over a fixed cut: a single fixed profile cut cannot be simultaneously near-optimal across the band and at C_5, whereas the *envelope* of per-root maxima equals d_{mono} at C_5 exactly. A relaxation built from a fixed convex combination of cuts is, by contrast, not a valid graphon upper bound; only the per-type minimum is — this is precisely the normalization ($\sum_c \lambda_{\sigma,c} = 1$ per type) that the earlier fixed-cut relaxation lacked.

Theorem 3.2. *There is an explicit nonnegative rational δ (numerically $\delta \approx 4.8558 \times 10^{-5}$; the exact value δ_{final} is recovered by the certificate re-verification of Section 6) such that*

$$d_{\mathrm{mono}}(W) \;\leq\; \tfrac{2}{25} + \delta$$

for every triangle-free $\{0,1\}$ step graphon W with $\mathrm{LO} \leq d_{\mathrm{edge}}(W) \leq \mathrm{HI}$, $\mathrm{LO} = \frac{1243}{5000} = 0.2486$, $\mathrm{HI} = \frac{3197}{10000} = 0.3197$. *Here δ is the optimum of an order-10 flag-algebra relaxation that* upper-bounds *the band maximum of d_{mono} (we never claim equality; only this upper bound is needed).*

It refines the order-9 per-root-MaxCut envelope of Lemma 3.1 ($d_{\mathrm{mono}} \le U_7$): besides the per-root rows at 7 roots it adds the per-root-MaxCut rows at 8 roots, rooted-Horn cuts, and a moment block, all of them valid for every triangle-free graphon.

The bound δ is the optimum of a linear program over the order-10 (C_5-lifted order-9) flag densities: a state vector $x \ge 0$ over the 12,172 triangle-free band states, $\sum x = 1$, with envelope variables u_σ. Its rows are (a) the band inequalities $\mathrm{LO} \le d_{\mathrm{edge}}(x) \le \mathrm{HI}$; (b) the per-root-MaxCut envelope rows at 7 roots (107 types) and at 8 roots (410 roots), each $u_\sigma \le g_{\sigma,c}(x)$ for a *fixed* Boolean rule c (so $u_\sigma \le \min_c g_{\sigma,c}(x)$), tight at the C_5-blow-up; (c) rooted-Horn cuts, valid global 2-colourings that vanish at every odd cycle; and (d) the moment-positivity rows $m(x) \ge 0$. Every genuine band graphon induces a feasible x, so the LP optimum δ *upper-bounds* the true band maximum of d_{mono}; the converse can fail — a feasible x need not come from any graphon — which is why the LP is a relaxation. By linear-programming duality, δ is certified by nonnegative multipliers: per-root envelope distributions $\lambda_{\sigma,c} \ge 0$ with $a_7 + a_8 = 1$, where $a_7 = \min_\sigma \sum_c \lambda_{\sigma,c}$ over the 7-root types and a_8 is the analogue over the 8-root roots; band multipliers $\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}} \ge 0$; and Horn and moment multipliers. These assemble into a single per-state dual-feasibility inequality $R_{\mathrm{cbh}}(s) \ge \langle Q, P(s)\rangle$ over all 12,172 states, where the moment term is governed by a *manifestly* positive-semidefinite Gram matrix $Q = \sum_c w_c\, vv_c\, vv_c^\top$ with $w_c \ge 0$ — so no semidefinite solve or Cholesky rounding is needed — and $\delta_{\mathrm{final}} = \rho + \mathrm{HI}\,\mu_{\mathrm{hi}} - \mathrm{LO}\,\mu_{\mathrm{lo}} - \frac{2}{25}a_8 + \varepsilon$, the slack ε absorbing an exact $O(10^{-8})$ per-state residual. This is the object re-verified exactly in Section 6.

4 The density tails

Theorem 4.1 ([2, Theorem 1.3])**.** *For N sufficiently large, every triangle-free graph G on N vertices with edge density at most* 0.2486*, and every such G with edge density at least* 0.3197*, satisfies $\beta(G) \le N^2/25$.*

Corollary 4.2. *For* every *N and every triangle-free G on N vertices whose* graphon *edge density d_{mono}-normalisation $2e(G)/N^2$ is at most* 0.2486 *or at least* 0.3197*, $\beta(G) \le N^2/25$.*

Proof. Apply Theorem 4.1 to the blow-ups $G[t]$. Each $G[t]$ is triangle-free on Nt vertices; in the convention of [2] its edge density is $e(G[t])/\binom{Nt}{2} = \big(2e(G)/N^2\big) \cdot \frac{Nt}{Nt-1} \to 2e(G)/N^2$ as $t \to \infty$. Hence for G with $2e(G)/N^2 < 0.2486$ (resp. > 0.3197 strictly), the blow-up density is eventually < 0.2486 (resp. > 0.3197), so Theorem 4.1 gives $\beta(G[t]) \le (Nt)^2/25$ for large t; by Lemma 2.1, $t^2\beta(G) \le t^2N^2/25$, whence $\beta(G) \le N^2/25$. The boundary values $2e(G)/N^2 \in \{0.2486, 0.3197\}$ lie in the *closed* certificate band of Theorem 3.2 and are covered there, so the band and the two tails cover all densities with no gap. □

5 Proof of Theorem 1.1

Let G be triangle-free on $N = 5n$ vertices, $1 \le n \le 40$, with monochromatic-pair density $d := 2e(G)/N^2$. We bound $\beta(G)$ by n^2 in all three density regimes; sharpness then follows from $\beta(C_5[n]) = n^2$.

Low tail ($d \le 0.2486$). By Corollary 4.2, $\beta(G) \le N^2/25 = n^2$.

Band ($0.2486 \le d \le 0.3197$*).* By Theorem 3.2 (whose band is closed) the limiting graphon W_G satisfies $d_{\mathrm{mono}}(W_G) \le \frac{2}{25} + \delta$, and $d_{\mathrm{mono}}(W_G) = 2\beta(G)/N^2$ by Section 2. Proposition 2.2 gives $\beta(G) \le n^2 + \frac{25}{2}n^2\delta < n^2 + 1$ (as $n \le 40$), so the integer $\beta(G)$ is at most n^2.

High tail ($d \geq 0.3197$). By Corollary 4.2, $\beta(G) \leq n^2$.

Every triangle-free G on $5n$ vertices, $n \leq 40$, therefore has $\beta(G) \leq n^2$, and $C_5[n]$ attains it. Hence $a(5n) = n^2$. □

6 Verification, certificates, and ground truth

The result rests on (a) the envelope soundness $d_{\mathrm{mono}} \leq U_7$ (Lemma 3.1) and (b) the validity of the order-10 certificate of Theorem 3.2. We guard both on independent levels; all numeric checks are in exact rational arithmetic (Python `fractions`), and the scripts are ancillary files.

(1) Envelope soundness $d_{\mathrm{mono}} \leq U_7$. Lemma 3.1 is a genuine graphon upper bound: each per-root rule defines one global 2-colouring, whose monochromatic density is at least d_{mono}, and U_7 is their type-averaged minimum. Unlike a bound obtained by averaging the bipartization of sampled finite subgraphs (which is invalid, since bipartization is non-local), no averaging trap arises. We confirmed $d_{\mathrm{mono}}(W) \leq U_7(W)$ by exhaustive enumeration of all triangle-free graphs of orders 6 and 7 (zero violations; equality only at C_5), and on a graph zoo: C_5 (ratio 1.000, tight), C_7 (1.002), C_9 (1.127), C_{11} (1.573), C_{13} (2.177), the Petersen graph (1.013), $C_5 \cup C_5$ (1.016). The envelope also survives the adversarial in-band weighted C_{11}-blow-up on which a fixed-cut relaxation fails: there $d_{\mathrm{mono}} = 8/2025 \approx 0.0040$ while $U_7 \approx 0.0108$.

(2) Certificate re-verification (two independent passes). The order-10 Horn certificate — the dual multipliers $(\rho, \mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}, \lambda_{\sigma,c})$ and the Gram weights w_c — is checked twice in exact `fractions`. First, the producer's verifier re-derives all five dual-feasibility conditions: $a_7 + a_8 = 1$; $\sum_c \lambda_{\sigma,c} \geq a_7$ for each of the 107 seven-root types; $\sum_c \lambda_{\sigma,c} \geq a_8$ for each of the 410 eight-root roots; the per-state inequality $R_{\mathrm{cbh}}(s) \geq \langle Q, P(s) \rangle$ for all 12,172 states (a worst-case residual -1.02×10^{-8} is absorbed by the exact slack ε); and $\delta_{\mathrm{final}} < 5 \times 10^{-5}$. Second, an *independent* gate re-derives $\delta_{\mathrm{final}} = 4.8557798 \times 10^{-5}$ *directly from the raw dual*, matching the producer's exact `Fraction` bit for bit, and confirms both $a_7 + a_8 = 1$ and $w_c \geq 0$ for every Gram atom. Since $\delta_{\mathrm{final}} < 2/(25 \cdot 1600) = 5.0 \times 10^{-5}$ this gives $n \leq 40$ with a 2.88% margin; the bound is *exact* (rational, not floating-point) and so carries no rounding risk. We do not claim $n = 41$: that needs $\delta < 4.7591 \times 10^{-5}$, which the non-deterministic moment LP does not robustly attain.

(3) Moment positivity (graphon-level, exactly certified). The moment rows enter as a sum-of-squares term with nonnegative multipliers; per flag the atoms are negative, and positivity is a graphon-level statement. For each root profile σ the averaged flag products factor through a Gram form, $\sum_H p_W(H) P^\sigma(H) = \rho_\sigma c_\sigma M^\sigma(W)$ with a positive scalar $\rho_\sigma c_\sigma$ and $M^\sigma(W) = \sum_c w_c q_c q_c^\top$ a sum of nonnegatively weighted ($w_c \geq 0$) rank-one terms, hence manifestly positive semidefinite — the exact rational form of Razborov's positivity theorem [5]. We verified each Gram form in exact rational arithmetic for all four profiles in use and for representative band graphons (the extremal C_5, in-band C_7, the Petersen graph, C_9, a weighted C_5): each is symmetric, matches an independent double-sum recomputation to exact-zero discrepancy, and gives the inner product against the certificate's positive-semidefinite matrix ≥ 0 exactly. The certificate's own moment term is governed by a second, manifestly positive-semidefinite Gram $Q = \sum_c w_c \, vv_c \, vv_c^\top$ with $w_c \geq 0$: these weights are found by a plain linear program over 394 exact atoms (77 in the support), so $Q \succeq 0$ holds by construction — no eigenvector recovery, semidefinite solve, or Cholesky rounding enters. The averaging is taken on the blow-up graphon (the same W_G as in Section 2); finite distinct-subset densities carry an $O(1/n)$ disjointness defect that does not affect the graphon bound.

(4) Brute-force ground truth. Independently of all flag machinery, we enumerated every triangle-free graph of order $9, 10, 11, 12$, computed β by exact maximum cut, and recorded d_{mono} for those in the band. The in-band maxima observed were 0.0494, 0.0400, 0.0496, 0.0556 respectively, all far below the certified $\frac{2}{25} + \delta \approx 0.0800$; no graph approaches, let alone violates, the bound. The max-cut routine was validated on known values ($C_5\colon 4$, Petersen: 12).

What is checked vs. cited. Levels (1)–(4) are computer-checked, the exact-arithmetic ones in rational arithmetic; the envelope soundness is an elementary graphon argument; the certificate's decision step (the LP optimum $\eta = \delta$ and its dual feasibility) is exact, with no `native_decide` and no floating-point threshold. The moment positivity rests on the cited theorem of [5], exhibited here as an exact Gram factorization. Theorem 4.1 is cited from [2].

7 The obstruction to all n

Theorem 1.1 stops at $n = 40$ because Proposition 2.2 needs $\frac{25}{2}n^2\delta < 1$, and the certified $\delta \approx 4.8558 \times 10^{-5}$ gives exactly $n \le 40$. A larger range needs a still smaller δ. The order-10 certificate is essentially at the flag-LP finite ceiling: the per-root-MaxCut envelope at 7 and 8 roots, even augmented with rooted-Horn rows and the moment block, plateaus near $\delta \approx 4.85 \times 10^{-5}$ (and its non-deterministic moment LP does not robustly reach the 4.76×10^{-5} that $n = 41$ would need), and order 11 is computationally out of reach. The sharp asymptotic conjecture needs $\delta = 0$ exactly, which no finite order of this relaxation delivers.

The natural route to all n is the asymptotic inequality $d_{\mathrm{mono}}(W) \le 2/25$ itself. It reduces (after fixing a maximum cut, with M the monochromatic edges and B the bipartite cut-graph) to a single *self-tight* statement on the invariant $\Gamma = \sum_{uv\in M}(d_B(u,v)+1)^2$: namely $\Gamma \le N^2$ when B is connected, which gives $\beta \le N^2/25$ since each bad edge has d_B even and ≥ 4 (triangle-freeness), so $\ell^2 \ge 25$. The per-component reduction and the single-block base case (sharp constant $25 = 5^2$ by AM–GM, tight at $C_5[q]$) are elementary; the open core is the connected-B transfer in the regime where the endpoints of a bad edge are joined by B-paths of different even lengths, and it is self-tight: every flag-algebra, signature, or quadratic-cut certificate we tried equals Γ at the $C_5[q]$ extremal, hence only re-proves $\Gamma \le \Gamma$. We regard isolating the conjecture to this single estimate, and establishing it computationally for the multiples of five up to $N = 200$, as the contribution of this note.

Ancillary files

`step1_v2_independent_gate.py` (the independent exact-`Fraction` verification: re-derives δ_{final} from the raw dual, and checks $a_7 + a_8 = 1$ and the manifest-PSD weights $w_c \ge 0$) and `complete_v2_cert.py` (the producer's one-command verifier of all five dual-feasibility conditions); `moment_gram_lp.py` and `moment_gram_exact_verify.py` (the manifest Gram block $Q = \sum_c w_c\, vv_c vv_c^\top$); `brute_dmono.py` with `flag_engine.py` (a self-contained independent brute-force max-cut ground-truth checker, requiring no certificate); the moment-PSD Gram-certificate scripts `g1_exact_psd.py`, `g1_graphon_density.py`; the compact dual (`horn_dual.pkl`, `moment_gram_w.pkl`, `v2_cert_complete.pkl`); a `README` giving the order parameter, the band endpoints, and the exact rational δ_{final}; and a `SHA256SUMS` manifest. The full order-10 LP cache and lifted-state tables exceed arXiv ancillary limits and are regenerable from the scripts.

References

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